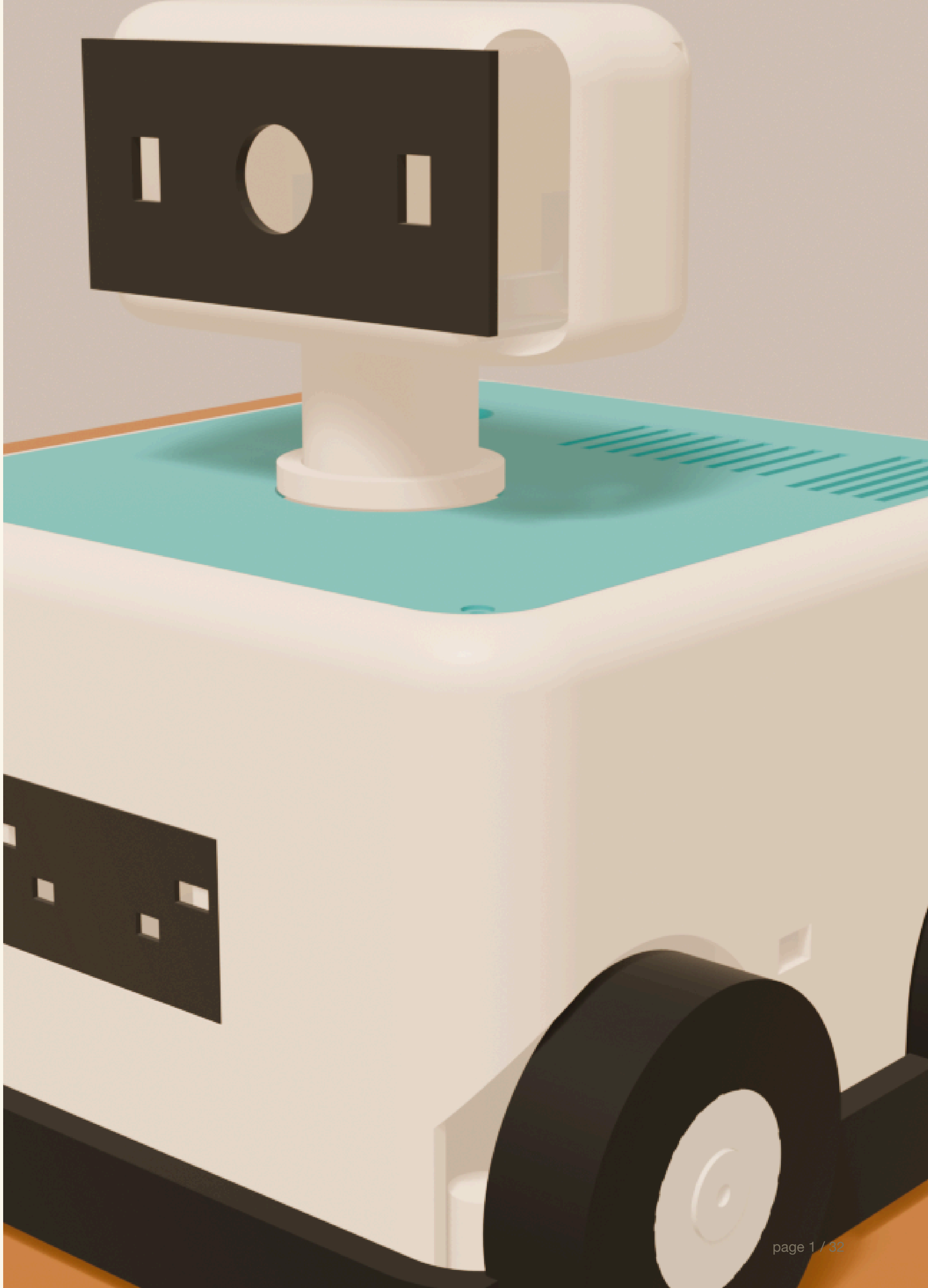
CODEX ROVER BEAN

The Robot Body Builder's Book

Print it. Screw it together. Meet your robot.
Everything you need is in this book and one Bambu Studio file.

AGES 10+ | 39 PRINTED PARTS | 13 PLATES | 1 HEX KEY | GROWN-UP FOR IRONS & WIRES



How this book works

The pictures do the talking

Every build step shows the robot so far, with the NEW parts floating just above where they land. Match the picture, then read the numbered lines if you want words too.

The parts strip

The tan strip at the top of each step shows exactly which parts and how many screws you need **before you start**. Lay them out like a cooking show.

The green CHECK bar

Do the little test at the bottom of each step before moving on. If the check fails, fix it now - later steps cover things up.

Red circles mean grown-up

Steps with a **red number** use the soldering iron or touch wires. A grown-up does those; you can watch and help.

One screw. One key.

Every screw in this robot is the same M3 x 8 screw, and one 2.5 mm hex key turns them all. "Snug" means: stop when it stops, then an eighth of a turn. Plastic hates gorillas.

Front and back

The FRONT is where the face panel and head look. The BACK has the charging plug, the mute switch, and the motor wheels. Left and right are the robot's left and right, not yours.

Chapter 1 - Go shopping: plastic, screws, tools

Filament (for the printer)

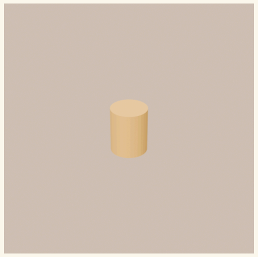
Filament	How much	What it becomes
Cream PETG	1x 1 kg spool (uses ~600 g)	The body, tray, head, and most parts.
Teal PETG	1x small spool or 250 g (uses ~80 g)	The top lid.
Charcoal / black PETG	leftovers are fine (~15 g)	Face panel, back panel.
Translucent lime PETG	a sample length (~5 g)	The glowing eye and light bars.
Charcoal TPU 95A	1x 500 g spool (uses ~280 g)	Soft tires, bumpers, battery pad.

The only two fastener packs

Fastener	Buy	Why
M3 x 8 mm socket head screws	1x 100-pack	The ONLY screw in the robot (40 used + spares).
M3 x 5.7 mm brass heat-set inserts (4.6 mm OD)	1x 100-pack	The only insert (40 used + spares).

Tools

Tool	Job
2.5 mm hex key	Turns every screw in this robot. Seriously, all of them.
Soldering iron (grown-up)	Melts the brass inserts into the plastic.
Small flush cutters / scissors	Trims zip ties and TPU strings.
Painter's tape + marker	Label wires as you go.



This screw and this brass insert are the only fasteners in the whole robot. When a step says "2 screws," it always means these.

Chapter 1 - Go shopping: the electronics box

Every item is a normal buy-one online part in the US. A grown-up orders these; exact fuse sizes and wire get picked in the wiring chapter.

Part	Qty	What it does
Raspberry Pi 5 (8 GB)	1	The robot's computer.
Raspberry Pi Camera Module 3 Wide + long FPC cable	1	The robot's eye.
Raspberry Pi Pico 2 (no headers, no WiFi)	1	The safety helper: reflexes and watchdog.
Cytron MDDS10 motor driver	1	The purple board that powers the wheels.
Pololu #4867 gearmotor (99:1, 25D, 12 V, encoder)	2	The wheel motors.
Hitec D85MG servo	2	The neck motors (look left/right, up/down).
Bioenno BLF-1203AB 12 V 3 Ah LiFePO4 battery	1	The robot's power pack.
Bioenno BPC-1502DC charger	1	The matching charger. Only ever use this one.
Switchcraft EN2P3M20 inlet + EN2C3F20G2 plug	1 pair	The keyed charging plug on the back.
Pololu D24V90F5 regulator (5 V)	1	Makes clean 5 V for the Pi.
Pololu D36V50F6 regulator (6 V)	1	Makes 6 V for the neck servos.
Panasonic CB1A-R-M-12V relay	1	The motor power switch the red button controls.
Blue Sea Systems 5045 fuse block + ATO fuses	1	Splits power safely into four fused branches.
IDEC XW1E-BV402M-R emergency stop	1	THE BIG RED BUTTON.
E-Switch PVB3F230SS311 mute switch	1	The microphone privacy switch (glows red when muted).
Omron D2HW-C202MR bumper switches	6	Feelers inside the bumpers.
VL53L1X time-of-flight boards (Adafruit 3967)	4	Distance eyes: two in front, one each side.
Adafruit 5975 NeoPixel breakouts + JST-SH cables	4	The glowing eyes and status lights.
ReSpeaker USB mic array	1	The robot's ears.
Enclosed 3 W 4 ohm speakers + 2x Adafruit MAX98357A amps	1 set	The robot's voice.
Wires, JST/spade connectors, ferrules, zip ties, heat-shrink	1 kit	Grown-up wiring supplies.

Chapter 2 - Print it with Bambu Studio

Open the project

Open **codex_robot_body_v2_p1s.3mf** in Bambu Studio. All 13 plates are already laid out for the P1S with a 0.4 mm nozzle and Textured PEI plate.

Load the right color

Each plate's name says the filament to load (cream, teal, charcoal, lime, or TPU). Print plates one at a time and change filament between groups.

No supports. Ever.

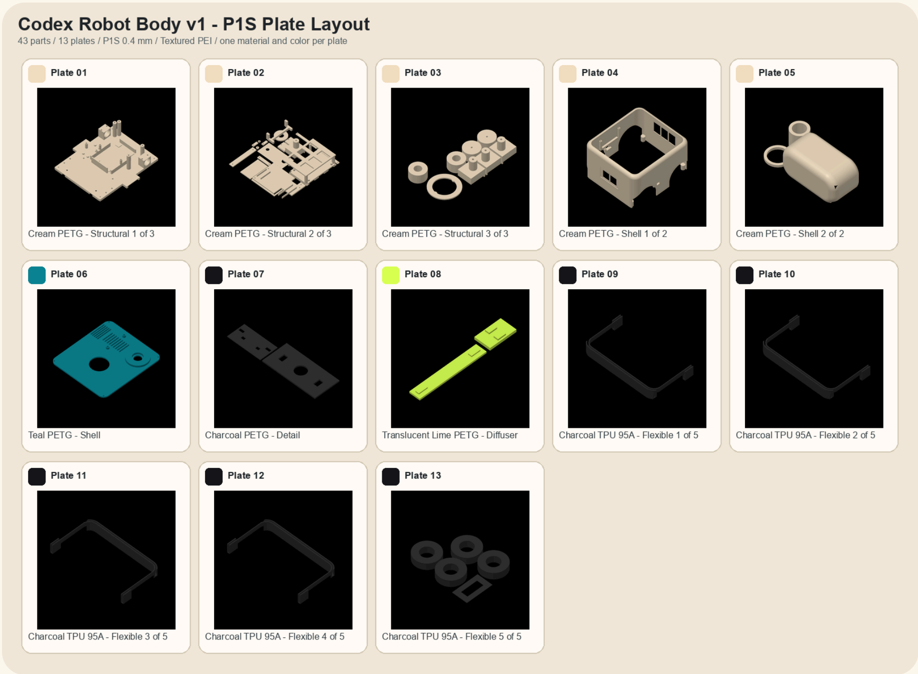
Every part is designed to print with zero supports. If the slicer asks for supports, something is wrong - don't add them, re-check the plate.

TPU is slow and squishy

Print the tire and bumper plates slowly (the profile already does this). Dry TPU prints much better.

Big flat parts stay put

The tray, shell, and bumper plates fill the whole bed. Clean the plate with dish soap first so they stick.



All 13 plates, exactly as they open in Bambu Studio.

Chapter 2 - The 13 plates, in printing order

#	Plate (load this filament)	Parts	What's on it
1	Cream PETG - Structural 1 of 3	1	tray
2	Cream PETG - Structural 2 of 3	16	deck, controller tower, rear panel, battery clamp, speaker clamp i1, speaker clamp i2, yoke, he
3	Cream PETG - Structural 3 of 3	8	mic cradle, front pod left i2, front pod right i1, front pod right i2, front wheel i1, front wh
4	Cream PETG - Shell 1 of 2	1	shell
5	Cream PETG - Shell 2 of 2	3	head shell, bayonet collar, neck
6	Teal PETG - Shell	1	lid
7	Charcoal PETG - Detail	2	head faceplate, fascia
8	Translucent Lime PETG - Diffuser	2	status diffuser bar, eye diffuser bar
9	Charcoal TPU 95A - Flexible 1 of 5	1	bumper front i1
10	Charcoal TPU 95A - Flexible 2 of 5	1	bumper front i2
11	Charcoal TPU 95A - Flexible 3 of 5	1	bumper rear i1
12	Charcoal TPU 95A - Flexible 4 of 5	1	bumper rear i2
13	Charcoal TPU 95A - Flexible 5 of 5	5	battery pad frame, tire i1, tire i2, tire i3, tire i4

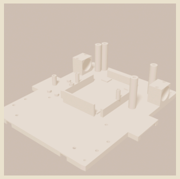
Tip: print plates 1-5 (cream) back to back, then change color once per group. Keep every part in a labeled box - the next chapter uses them in order.

Chapter 2 - Print the little test parts FIRST

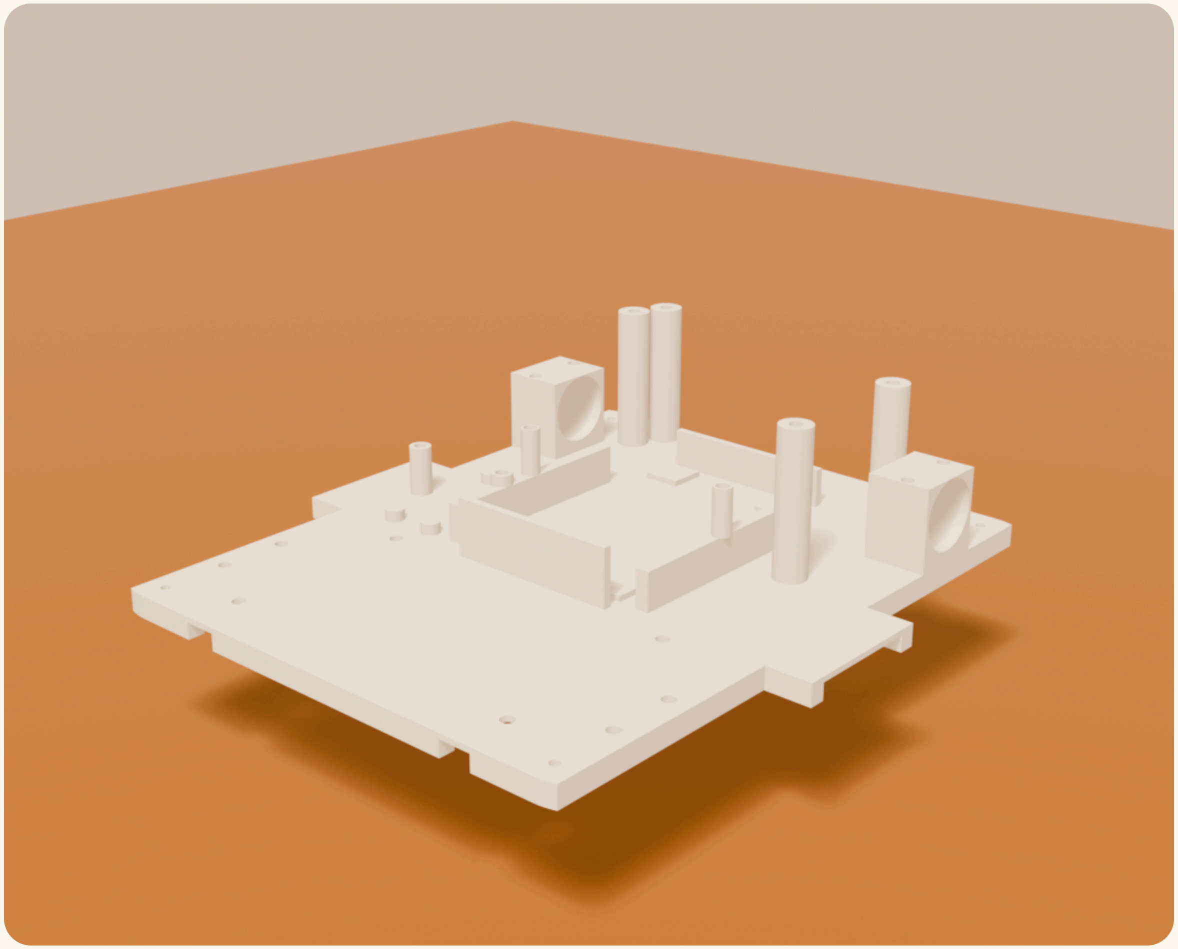
Before the big plates, print these small test parts (in **cad/exports/v2/coupons**). They make sure your printer's holes, snaps, and fits are dialed in - like tasting the batter before baking the whole cake. A grown-up checks each one.

Test part	Filament	What it proves
insert m3	PETG	insert fit: drive one insert per bore, pick the station that seats flush without melt squeeze-out
joint flange	PETG	with coupon_joint_boss: M3x8 through the counterbored flange; verify 3.2 clamp, full seat, no bottoming
joint boss	PETG	mate of coupon_joint_flange
dbore torque	PETG	mount on the #4867 shaft: hold 2x extrapolated stall (22 kg-cm) without slip or creep, warm and cold (D025 gate)
axle stub	PETG	with coupon_bushing_ring: spin under ~1 kg radial load; measure slop growth over a carpet-equivalent drag distance (D025 gate)
bushing ring	PETG	wear ring for coupon_axle_stub; grease PETG-safe
snap pair	PETG	10 insert/release cycles; hook must survive and retention must stay positive
bayonet pair	PETG	head-retention interface: engage/disengage feel, no ratcheting under axial pull
switch pocket	PETG	seat one D2HW switch; verify the 0.4 rest gap / 2.0 stroke / 2.4 stop interface before committing the tray print
tire fit	TPU+PETG	print the ring in TPU, the core in PETG; calibrate tire stretch and seat retention at 40% scale
pcb clamp	PETG	drop in the Pico 2, verify peg engagement without board stress, clamp bar seats at 3.2 stack

1 Start with the floor



x1

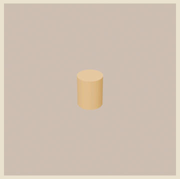


1. Clear a big table. Put the tray down flat, wheels-cutouts toward you.
2. Find the two round motor cradles at the back and the tall round towers - that's the back of the robot.

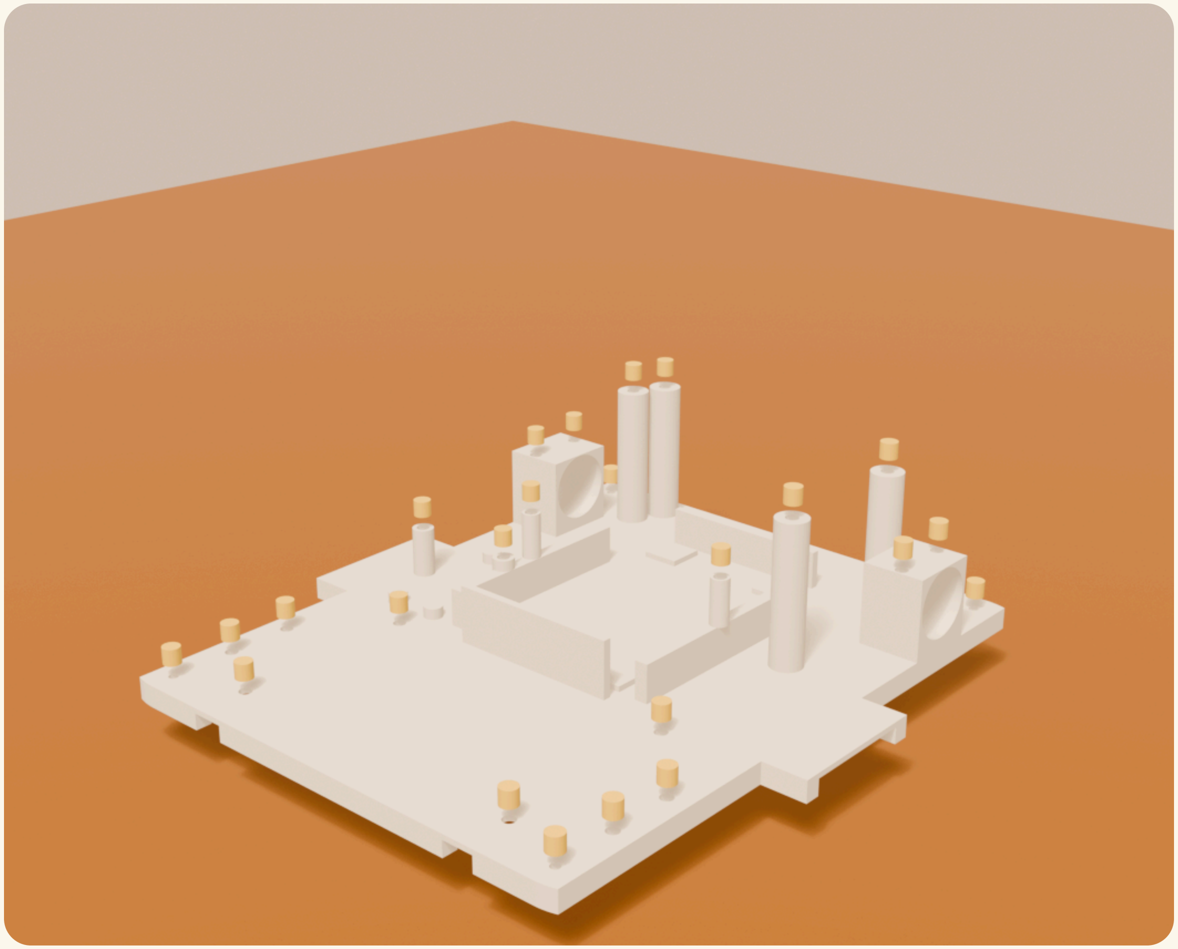
CHECK The tray sits flat and doesn't rock.

GROWN-UP STEP

2 Melt in the brass inserts



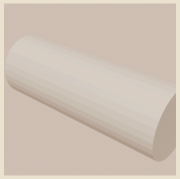
x24

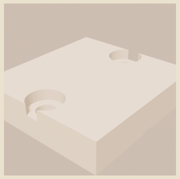


1. GROWN-UP: set the soldering iron to about 220 C.
2. Rest a brass insert in each gold-marked hole, then press it straight down with the hot iron tip until it sits flush.
3. Let each one cool. The shell, saddle tops, rails, and towers all get inserts too - the picture shows every spot on the tray.

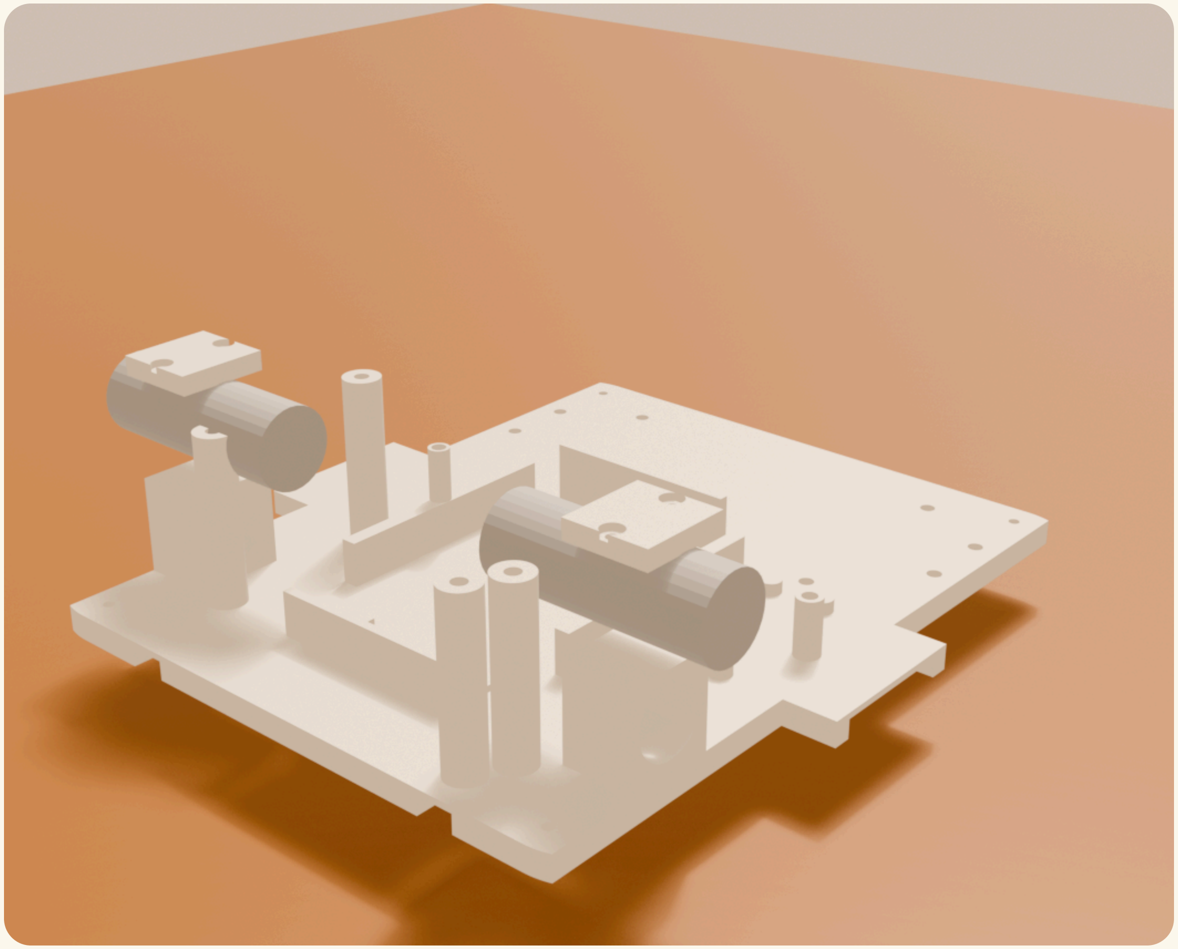
CHECK Every insert is flush and straight, none tilted.

3 Drop in the motors


x2


x2

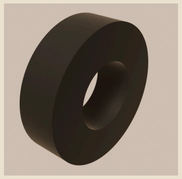

x4
M3 screws

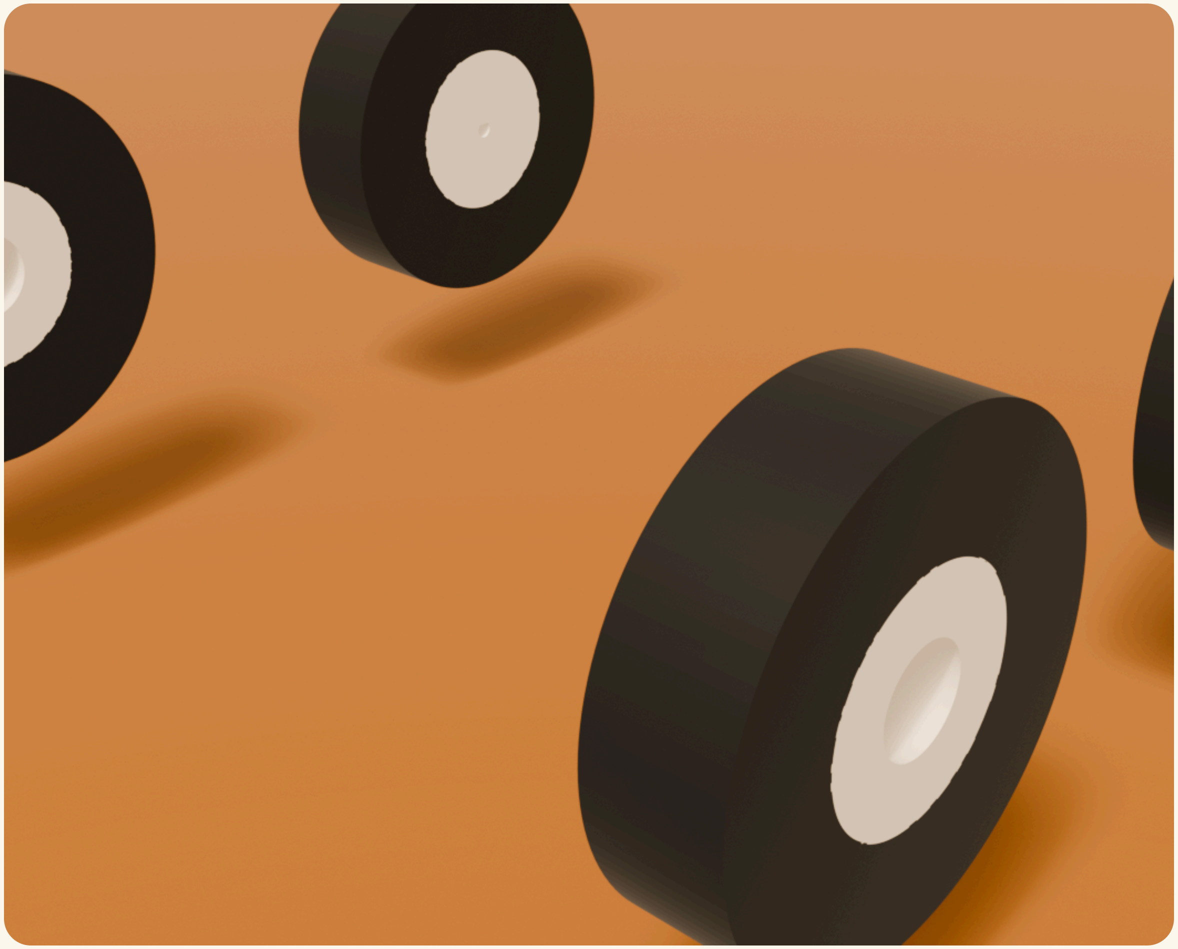


1. Lay a motor in each cradle with the metal shaft poking OUT through the hole toward the wheel side.
2. Point the wires inward, toward the middle of the robot.
3. Set a printed cap over each motor and screw it down with 2 screws per cap - snug, not gorilla-tight.

CHECK Motors don't wiggle. Shafts spin freely when you twist them.

4 Make the wheels

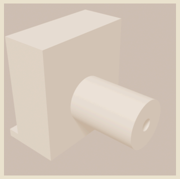
 x2  x2  x4



1. Stretch a rubber tire over each of the four wheels, like putting a rubber band on a yo-yo.
2. Work it around evenly until it sits flat in the groove all the way around.

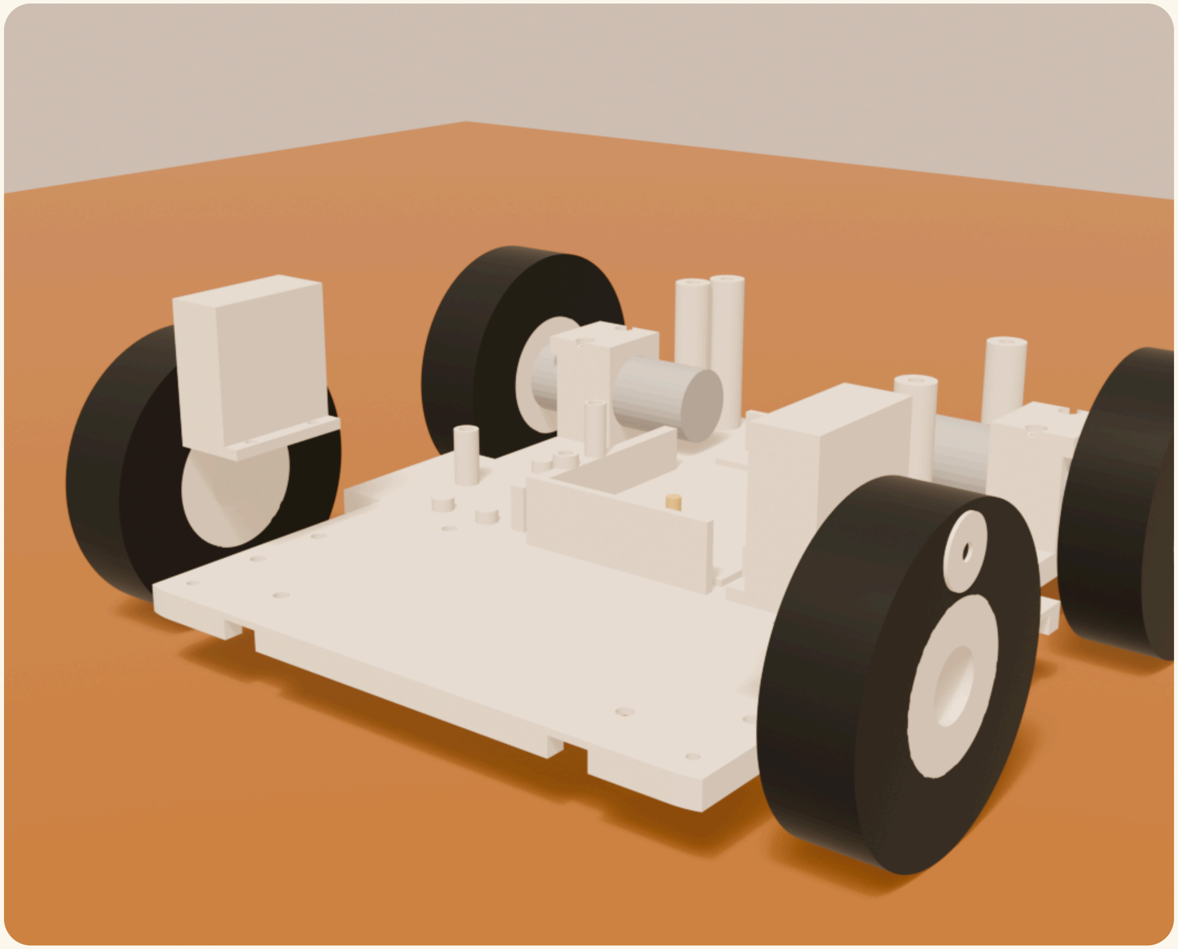
CHECK **No tire bulges. All four look the same.**

5 Bolt on the front legs



x2

x4
M3 screws



1. The two front pods have round pegs sticking out - those pegs are the front axles.
2. Screw each pod to the tray through its little foot tabs, 2 screws each, pegs pointing OUT.

CHECK Both pegs point straight out to the sides.

6 Put the wheels on

x2

x2

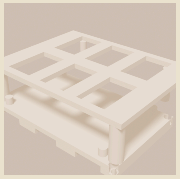
x2

x4
M3 screws

1. BACK wheels: the hole has a flat side, and so does the motor shaft. Line the flats up, push the wheel on, then tighten the one clamp screw on the wheel's rim.
2. FRONT wheels: slide onto the pegs - they should spin freely. Put a printed washer on, then a screw into the end of the peg to keep the wheel from sliding off.
3. Don't overtighten the front screws: the wheels must still spin.

CHECK Back wheels turn only when the motor turns. Front wheels spin freely.

7 Build the brain tower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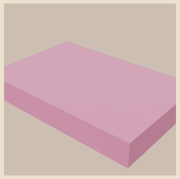
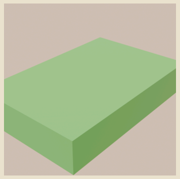

x1


x4
M3 screws

1. Set the tower over the front-left of the tray - its screw holes match the four inserts.
2. Drive 4 screws down through the base tabs.
3. Rest the purple motor board on the four little posts, its green terminal blocks facing the LEFT side of the robot.

CHECK The board sits level on all four posts, terminals facing left.

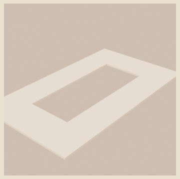
8 Add the computer

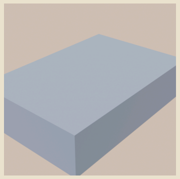
 x1

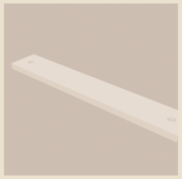
1. The Raspberry Pi lies flat on the tower's top shelf frame.
2. Its USB ports face the BACK of the robot so the cables can reach.
3. Don't screw anything - the shelf pocket holds it, and the head's cable will come down to it later.

CHECK The Pi sits in its pocket, ports facing backward.

9 Strap in the battery

 x1

 x1

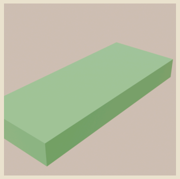
 x1

 **x2**
M3 screws

1. Lay the soft TPU pad frame onto the four pads behind the tower.
2. Set the battery on it, wires pointing at the BACK of the robot.
3. Bridge the clamp bar across the battery onto the two posts and screw it down with 2 screws - firm, so the battery cannot slide.

CHECK Grab the battery and try to wiggle it. It shouldn't move.

10 Add the safety helper



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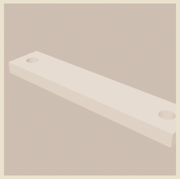


x1

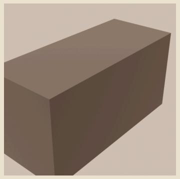
x2
M3 screws

1. The tiny green Pico sits on its little posts to the right of the tower, USB plug facing RIGHT.
2. Lay the small clamp bar across it and screw it down with 2 screws, gently - it's a small board.

CHECK The Pico is held snug and its USB port is reachable.

11 Mount the power relay

GROWN-UP STEP

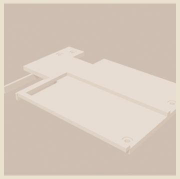


x1

1. GROWN-UP: the relay drops into its floor pocket on the left, behind the tower.
2. Its metal bracket slots into the printed pocket; the terminals face UP so you can wire them later.
3. No wires yet - all wiring happens in the wiring chapter at the back of this book.

CHECK The relay clicks into its pocket and doesn't rattle.

12 Put on the power deck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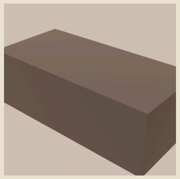
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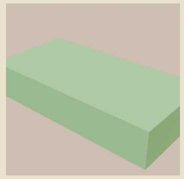
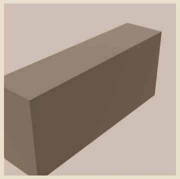
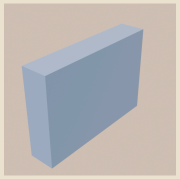
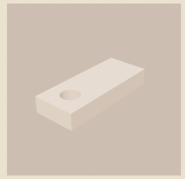
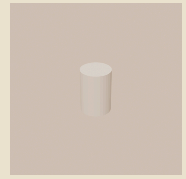

x2


x4
M3 screws

1. First hang the two small green regulator boards under the deck's east end (they clip under; wires come later).
2. Lower the deck onto the four towers - the notch at the back-right corner goes around the battery wires.
3. Drive 4 screws down into the tower tops.
4. Set the black fuse box into its raised outline on the deck's right half, wire end hanging over the edge.

CHECK The deck is level and the fuse box sits inside its printed fence.

13 Speakers and distance eyes

 x2  x2  x2  x2  x6 M3 screws

1. Rest a speaker on each side shelf inside the shell area, magnet side in, grille facing the wall.
2. Lay a clamp bar across each speaker's top and screw into the two posts (2 screws per side).
3. Slide a little blue distance board behind each side window, then its clamp bar and 1 screw.

CHECK Speakers can't rattle; the blue boards peek through their windows.

14 Lower the body shell

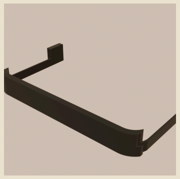
x1

x4
M3 screws

1. Two people make this easy: lower the big shell straight down over EVERYTHING.
2. The wheel arches go around the wheels; the lip settles onto the tray edge.
3. Flip-check the underside: drive 4 screws UP through the tray's corner holes into the shell's lugs.

CHECK No gaps between shell and tray. The robot is now a box with wheels.

15 Clip on the bumpers

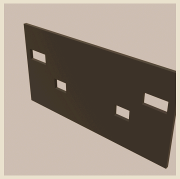


x2

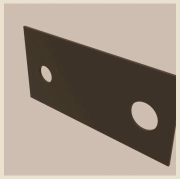
1. The two soft bumper halves wrap around the bottom, one from the front, one from the back.
2. They hug the body loosely on purpose - they need to squish in to press the hidden feeler switches.

CHECK Press any bumper edge gently: it moves in a tiny bit and springs back.

16 Snap in the face and back panels



x1



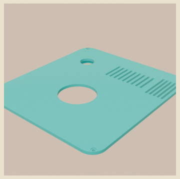
x1

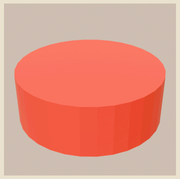
1. The dark FACE panel clicks into the front opening (the two sensor holes go low).
2. The dark BACK panel clicks into the back opening - its two round holes are for the charger plug and the mute switch, wired later.

CHECK Both panels sit flush with the body.

17 The lid and the BIG RED BUTTON

GROWN-UP STEP

 **x1**

 **x1**

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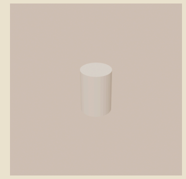
 **x6** M3 screws

1. GROWN-UP: drop the red emergency-stop through the lid's round hole and spin its nut on underneath - the lid IS its mounting panel, and the printed ring under the lid makes it strong.
2. Set the round microphone under the lid's slotted area and screw its ring cradle to the two bosses (2 screws).
3. Wires come later. Lower the lid into its ledge and drive the 4 corner screws.

CHECK Slap test: the red button clicks down hard and twists to release.

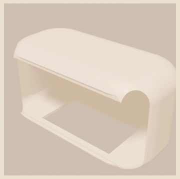
18 Grow the neck

 x1  x1  x1

1. Feed the neck tube down through the lid's front hole.
2. From inside, twist the bayonet collar onto the neck's bottom - a quarter turn locks it, like a camera lens.
3. The neck servo sits in the collar's cradle underneath (its wire joins the wiring chapter).

CHECK The neck turns smoothly by hand and cannot pull up and out.

19 Build the head

 x1

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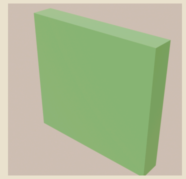 x1

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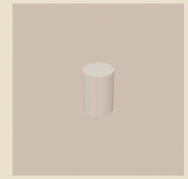
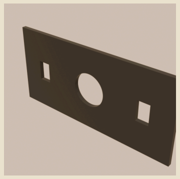
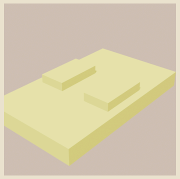
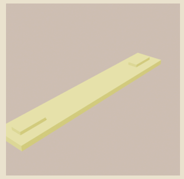
 x1
M3 screws

1. Screw the yoke's ring onto the top of the neck.
2. Slide the camera into the pocket behind the face opening, lens forward; its flat ribbon cable runs down through the hollow neck to the Pi.
3. Lower the head shell over the yoke: one side takes the tilt servo, the other side gets the little bushing and 1 pivot screw.
4. Close the underside with the pan plate.

CHECK The head nods up-down and the neck turns left-right, smoothly.

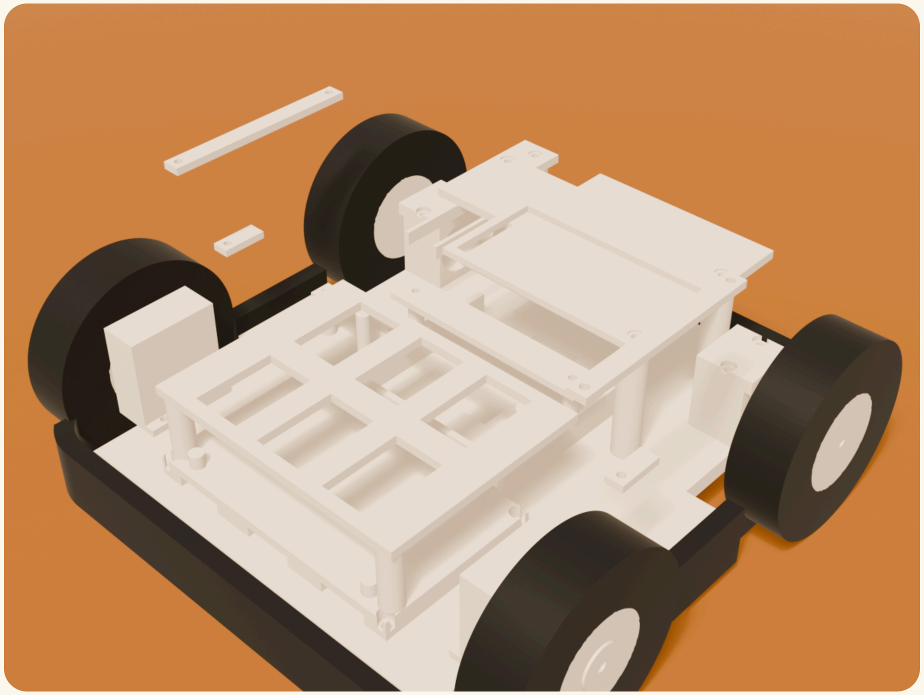
20 Give it a face

 x1  x1  x1

1. Clip the lime eye bar behind the face panel's eye slots (glow boards ride behind it).
2. Press the face panel into its recess: camera hole over the lens.
3. The second lime bar clips behind the front body panel's light slots.

CHECK **Rover Bean is looking at you. Say hi.**

Chapter 4 - Where every electronic part lives



Part	Home	Faces
MDDS10 motor board	tower posts, front-left	terminals LEFT
Raspberry Pi 5	tower top shelf	USB ports BACK
Pico 2	floor, right of tower	USB RIGHT, pins UP
Battery	middle-back floor	wires BACK
Relay	floor pocket, left	terminals UP
Fuse block	deck, right half	wire exit RIGHT edge
Regulators (5 V + 6 V)	hang UNDER deck, back end	6 V wire exit LEFT
Big red button	through the lid	UP (slap it!)
Mic array	under the lid slots	UP
Speakers	side shelves	grilles OUT
ToF distance boards	2 behind face, 2 side windows	lenses OUT
Charger inlet + mute	back panel holes	BACK

The two speaker amp boards (MAX98357A) zip-tie beside their speakers for now - printed pockets for them are on the upgrade list.

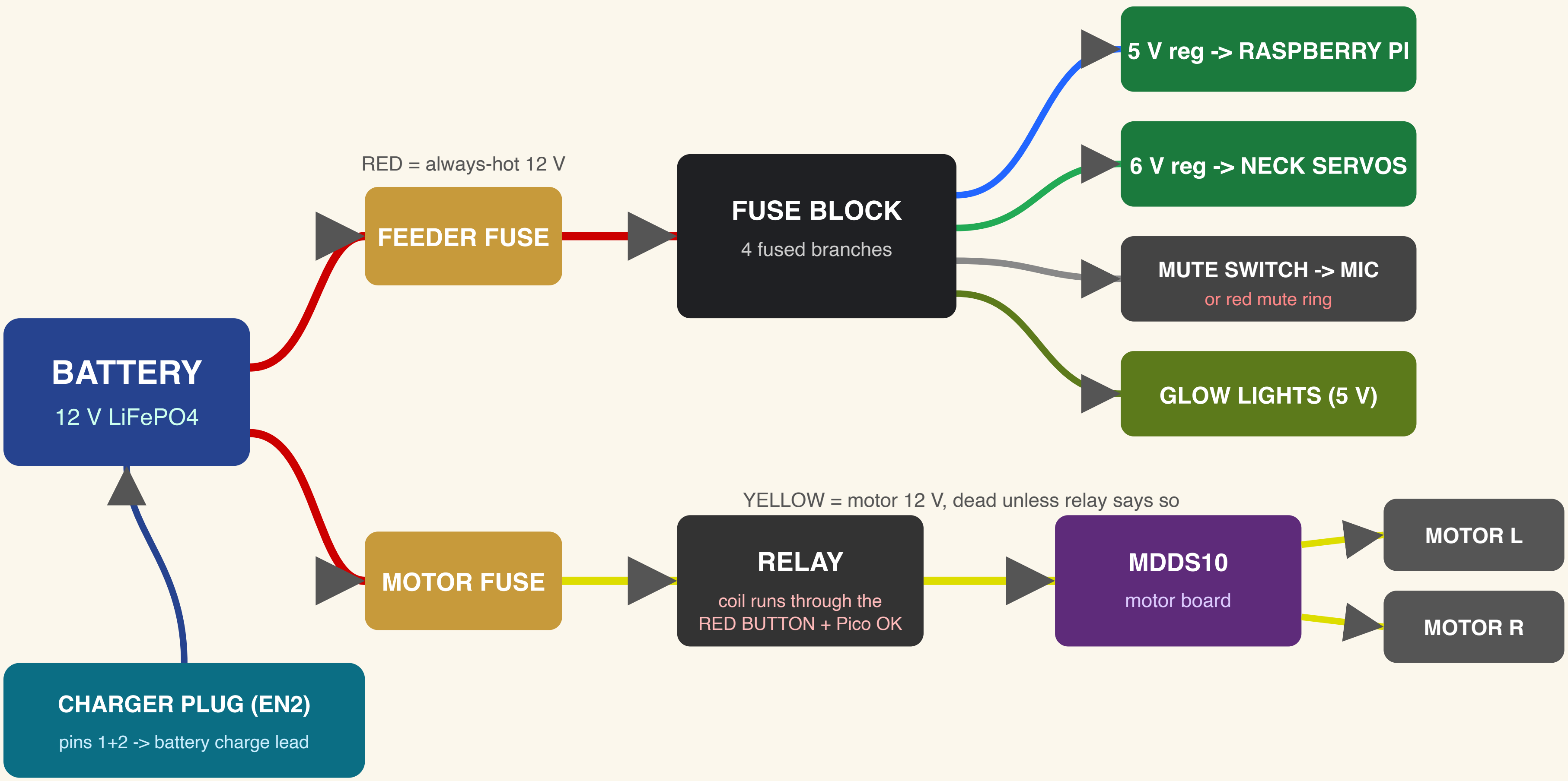
Chapter 4 - Grown-up wiring rules

- This whole chapter is grown-up work. The builder can watch and hand you zip ties.
- The battery stays OUT of the robot until every wire is checked against these maps.
- Use wire colors: RED = battery 12 V, YELLOW = switched motor 12 V, BLUE = 5 V, GREEN = 6 V, BLACK = ground, WHITE = signals.
- Crimp or solder every joint; no bare twists. Label both ends of every wire with tape.
- Fuses go in LAST, after a meter check - start with the smallest sizes that hold.
- The robot must FAIL STOPPED: if any of this feels wrong, it stays off.

The power map (every 12 V wire)

From	Through	To
Battery +12 V	feeder fuse (near battery)	Blue Sea fuse block IN
Fuse branch 1	D24V90F5 regulator	5 V to the Raspberry Pi
Fuse branch 2	D36V50F6 regulator	6 V to both neck servos
Fuse branch 3	mute switch common	mic USB power OR red mute ring
Fuse branch 4	5 V accessories	NeoPixel eyes + status lights
Battery +12 V	motor fuse -> relay contacts	MDDS10 motor board power
Relay coil 12 V	through BOTH red-button NC contacts	coil ground via Pico enable
Charger EN2 pins 1+2	direct to battery charge lead	pin 3 = charger-present to Pico

Chapter 4 - Power, as a picture



Chapter 4 - The signal map (thin wires)

From	To	What travels
Raspberry Pi	MDDS10	serial motor commands (through the Pico's watchful eye)
Raspberry Pi	Pico 2	USB: heartbeat + status; Pico can always stop motors
Motor encoders (6 wires each)	Pico 2	wheel speed feedback
Bumper switches x6	Pico 2	any press = stop, instantly, no software needed
ToF boards x4	Raspberry Pi	I2C daisy chain (STEMMA cables)
NeoPixels x4	Raspberry Pi	one data line, chained eye->eye->status->status
Camera	Raspberry Pi	flat FPC ribbon down the hollow neck
Servos x2	Raspberry Pi	PWM signal wires (power from the 6 V rail)
Mic array	Raspberry Pi	USB (its 5 V passes through the mute switch)
Speakers	MAX98357A amps -> Pi I2S	amps zip-tie beside each speaker for now

Cable neatness

Quiet wires (signals, audio, sensors) ride the LEFT channel under the deck. Power wires ride the RIGHT channel. The camera ribbon goes up the little tunnel beside the neck servo, through the hollow neck, into the head. Zip-tie at every printed tie point; no wire may touch a wheel or the fan.

Before the battery goes in - the grown-up meter checklist

1) No shorts: beep-test + to - at the fuse block (no fuses in yet). 2) Red button held down = relay coil circuit OPEN. 3) Regulator outputs read 5.0-5.2 V and 6.0 V on the bench before their loads connect. 4) Fuses in smallest-first, one branch at a time. 5) Charger plug in = motors will not run. Only then does the lid close.

You built a robot!

High five, builder. Rover Bean's body is done.

Chapter 5 - Before it ever moves (grown-ups)

Slap the red button while the wheels spin on blocks - everything must stop instantly. Press each bumper - stop. Unplug the Pi's heartbeat - stop. Plug in the charger - it refuses to drive. Only after every one of these passes does Rover Bean get floor time, supervised, at walking pace.

The robot's number one rule, forever: **when anything is wrong, it stops.**

Codex Rover Bean v2 - printed-only body - one screw, one key, zero supports.